Sandboxing and resource limits

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Check the scope. Check every quota. Require enforcement evidence.

Evaluate declared sandbox/resource policies without confusing pure calculations with real isolation or output correctness.

Keep one habit: Main scope/admission contract → operation map → queue boundaries → separate trial picture → quota/enforcement probes → local policy audit → independent queue case.

MAIN LESSON · SECTION 1

Check what a job may reach and how much it may consume

> Permissions, time, output and queue limits are separate conditions.

Learning goals

I can explain the **mechanism** and terminology of **sandbox**ing and resource limits.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve exact input and trusted action boundaries

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Untrusted **input** boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Untrusted **input** boundaries. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**
>
> _____
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> _____
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> _____
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> _____

End goal

Explain **sandbox**ing and resource limits using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of **sandbox**ing and resource limits

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Declare reach and quotas before accepting work

> A policy calculation is a model of required enforcement, not the enforcement itself.

Begin with the school lab’s bounded job

A school lab can allow a job to read its fixture area and write only its output area, while giving it no network access. The fact that a webpage is public does not add network permission. Separately, the lab can limit elapsed time and queued work. A job must satisfy every applicable condition.

We will inspect declared trial records and calculate policy decisions; the local Python exercise does not isolate or execute a real student program.

Define sandboxing and resource quotas

Sandboxing and **quota**s constrain what a **process** may access or consume. They reduce possible operations but do not verify every result.

Complete main scope and admission contract

Main fictional job policy	Complete meaning in the local model
Read scope	Fixture area only, represented by abstract handle fixtures

Main fictional job policy	Complete meaning in the local model
Write scope	Output area only, represented by abstract handle outputs
Network	No network permission, even for public information
Time cap	Declared elapsed-runtime maximum 2 seconds; actual enforcement not implemented
Queued-task cap	At most 10 pending tasks
Admission policy used here	All-or-nothing batch: reject if current + proposed exceeds 10
Code representation	Exact operation/resource pairs and arithmetic checks, not filesystem paths
Observation	Permission decisions and supplied quota records, not actual network/file effects
Separate picture	Adds its own 20 MB output cap; do not silently borrow it as a main-policy fact

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Separate requested-operation decisions from queue admission

> Passing one resource condition cannot cancel another prohibition.

Original read/write/network/time/queue mechanism

The fictional job may read its fixture directory, write its **output** directory, use no network and run for at most 2 seconds with at most 10 queued tasks. Reject excess work before starting it.

Complete requested-operation trace

Check each requested operation against its own scope

REQUESTED OPERATION	POLICY CHECK	DECISION
Read fixtures	Exact declared read/resource pair	Permitted in the policy model
Write outputs	Exact declared write/resource pair	Permitted in the policy model
Fetch public webpage	No network capability	Deny before network execution
Write outside output area	Resource outside write scope	Deny before filesystem effect
Write fixture area	Read permission does not imply write permission	Deny undeclared pair

Abstract handles represent the scope decisions. This is not a path-prefix check, filesystem sandbox or proof that a real process cannot escape.

Explicit queue boundaries and unchanged-state policy

Main admission fixture	Projected queued tasks	All-or-nothing decision
8 currently queued +2 proposed	10	Accept batch in the policy calculation
8 currently queued +3 proposed	11	Reject batch; current queue remains unchanged in the model
Different partial-admission policy	Requires explicit separate rule	Do not silently choose a partial batch
Concurrent real producers	Need actual coordinated admission	Pure arithmetic does not atomically reserve capacity

State limits and units before checking

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Sandboxing and resource limits — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A job requests a public webpage and a write outside its output directory.	The fictional job may read its fixture directory, write its output directory, use no network and run for at most 2 seconds with at most 10 queued tasks. Reject excess work before starting it.	Both operations violate the sandbox contract. Deny them; public availability does not grant network or filesystem authority.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change one resource record while holding the others fixed

> A small output does not excuse a time-limit failure.

Current task and response

- A job requests a public webpage and a write outside its **output** directory. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**
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> _____
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> _____
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> _____

Predict the three-second pictured trial

- Change: A takes three seconds but still writes 2 MB.

▼ **Check after predicting**

Prediction: Stop A for exceeding the time allowance.

Reason: The independent two-second limit remains binding.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Check the pictured time, output and network conditions independently

> These are declared trial records, not programs executed in a real sandbox.

Main requested public-web/outside-write actions and denial

A job requests a public webpage and a write outside its **output** directory.

Both operations violate the **sandbox** contract. Deny them; public availability does not grant network or filesystem authority.

Keep the reasoning's scope

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate picture caps, units and supplied trial records

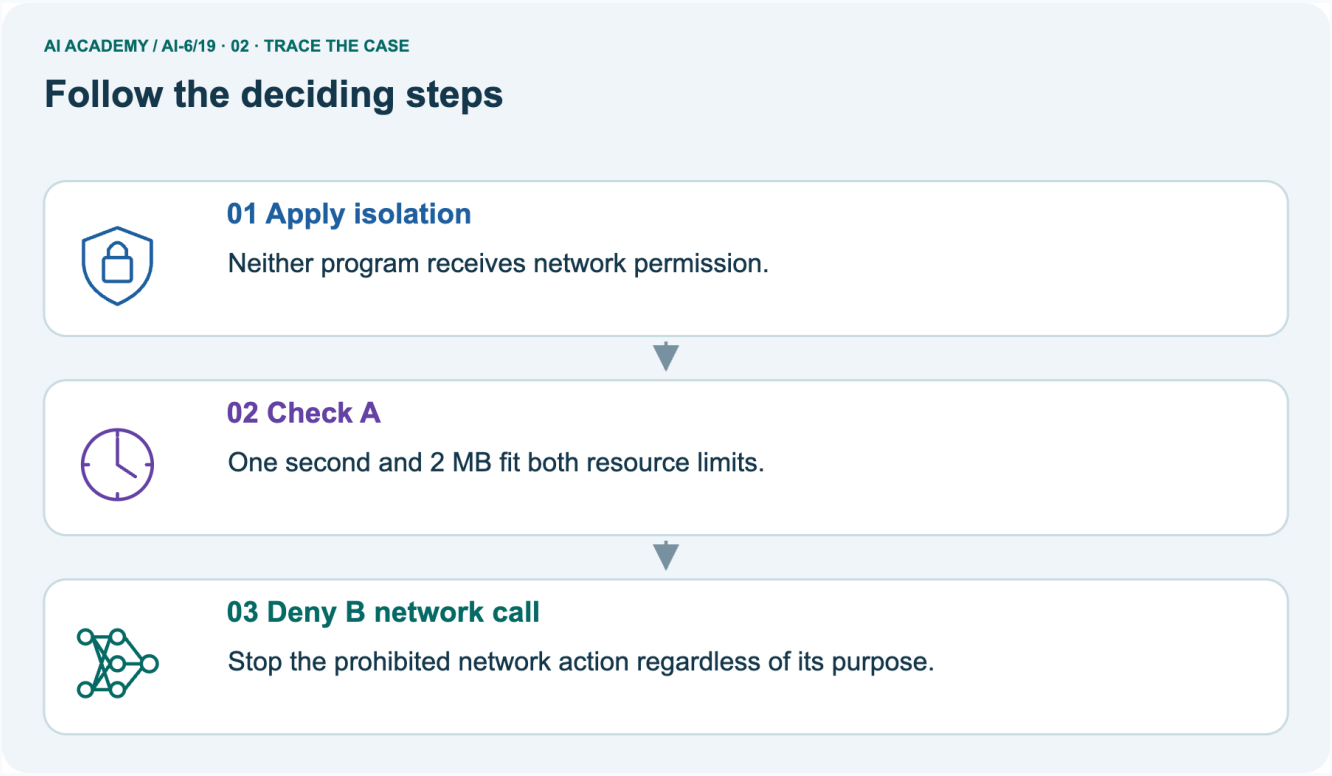
Bound a student program. A toy **sandbox** forbids network access, allows at most two seconds and limits **output** to 20 MB.

A uses one second and writes 2 MB. B tries to contact a website. These are declared trial records.

The main policy includes its queue cap; the picture is a separate policy with no network access, at most 2 seconds and at most 20 MB of output. Here 1 MB means one million bytes. Supplied trial A takes 1 second and writes 2 MB, so those resource records fit. Trial B proposes a network call, which must be denied regardless of public availability.

A changed record taking 3 seconds exceeds the independent 2-second limit even with only 2 MB of output. A prohibited attempt does not prove that a real packet was sent; this exercise specifies a required policy decision, not an observed sandbox escape.

Trace fitting resources and prohibited network request



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Apply isolation	Neither program receives network permission.
2. Check A	One second and 2 MB fit both resource limits.
3. Deny B network call	Stop the prohibited network action regardless of its purpose.

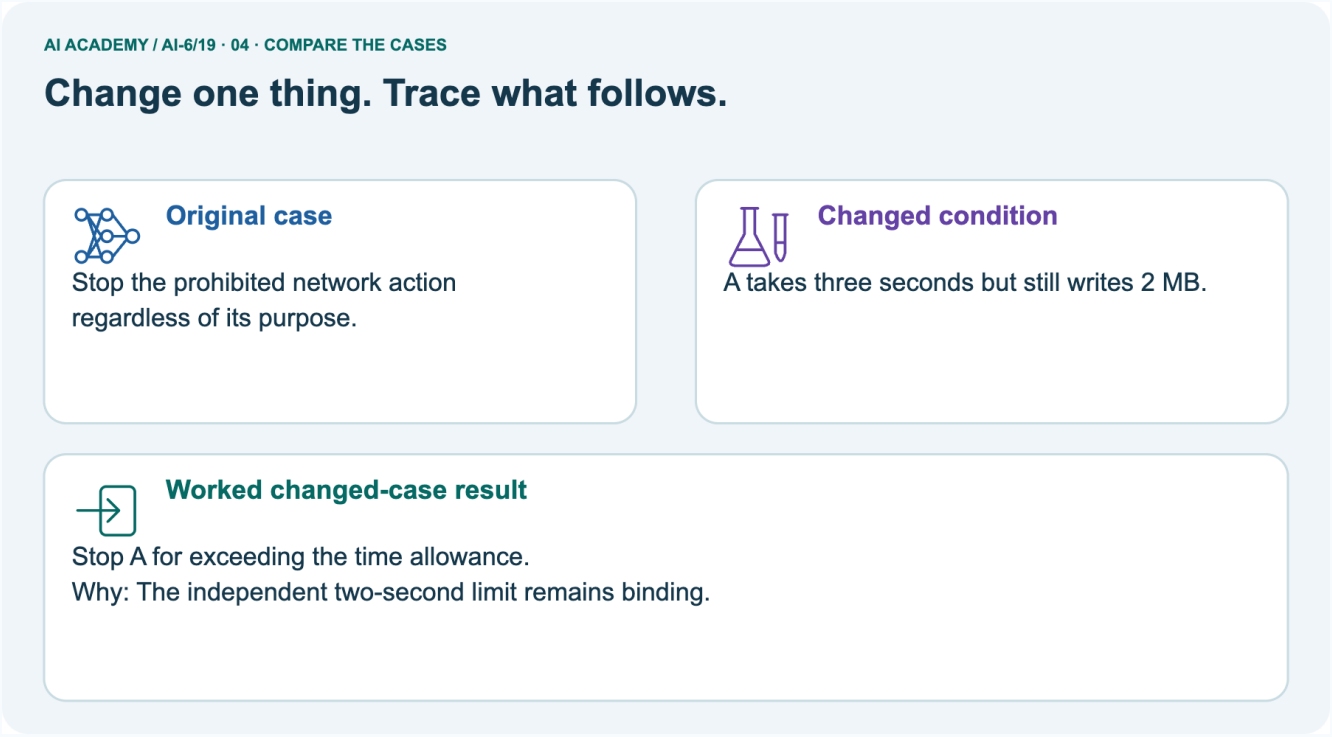
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Sandbox controls bound reach and resource use. Passing an **output** limit cannot cancel a network or time restriction.

Predict A taking 3 seconds with 2 MB output

- Change one thing: A takes three seconds but still writes 2 MB. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain independent limits and enforcement evidence still missing

- These controls are not a complete security proof; implementation and escape risks still require review.

- Read the labelled picture: Check A against both resource limits and B against network isolation. Why must A stop after three seconds even if its **output** remains only 2 MB?

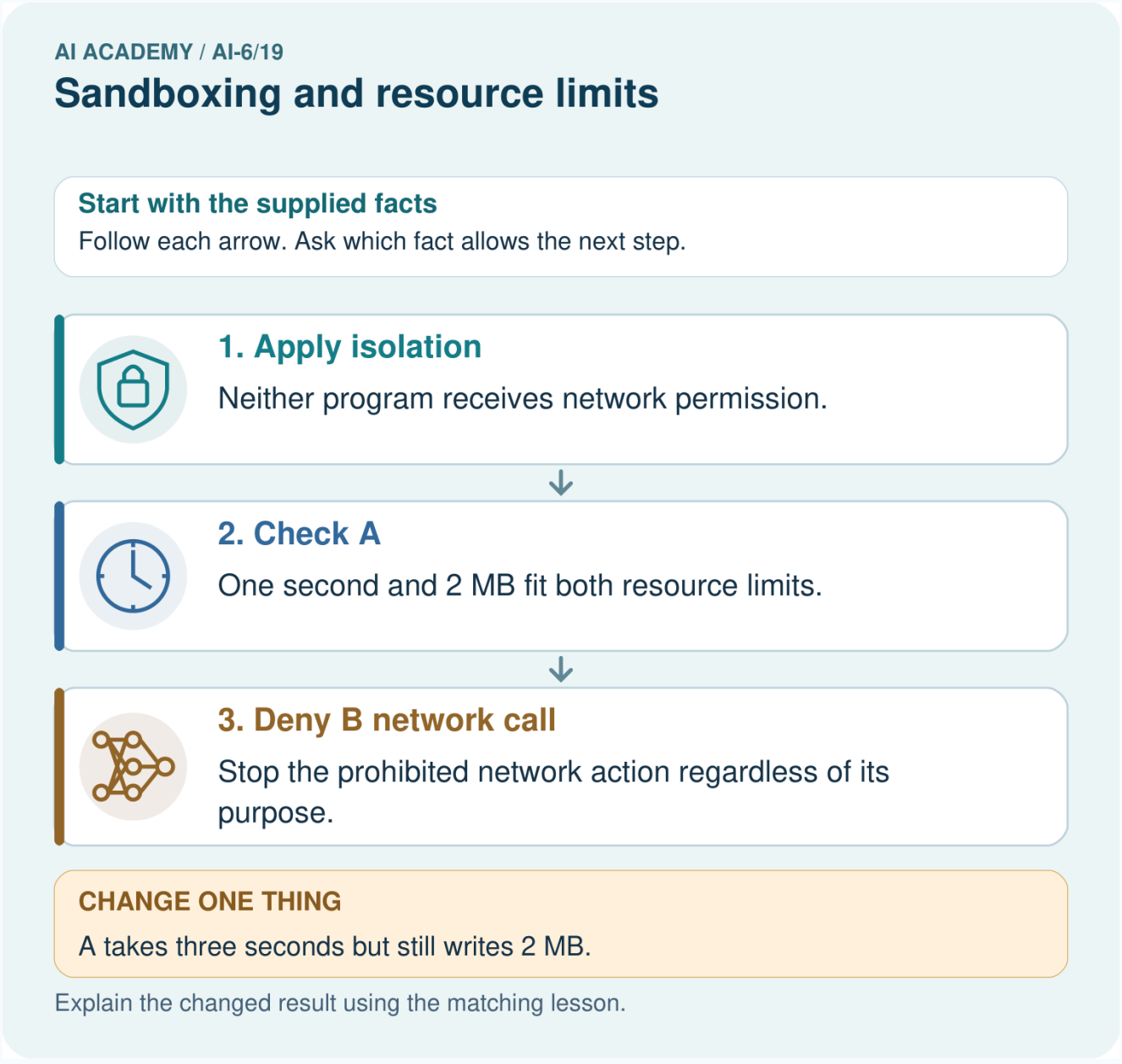
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sandboxing and resource limits

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Know what compliance with a quota actually establishes

> Resource control does not make a generated claim or algorithm correct.

Core resource-versus-correctness qualification

Resource limits constrain execution but cannot prove generated claims or algorithms correct.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete scope, unit, boundary and implementation-limit probes

Policy or trial probe	Expected interpretation
Public webpage requested with no network scope	Deny; public availability is not capability authority
Write outside output area	Deny; unrelated allowed resource use does not permit it
Pictured 2 seconds and 20 MBexactly, no network request	Fits inclusive maxima in the supplied-record calculation
Pictured 20 MB plus 1 byte	Output-limit finding even when time fits
Pictured 3 seconds with 2 MB	Time-limit finding even when output fits
Queue projection exceeds maximum	Reject excess batch before admission under the declared policy
Negative/nonfinite duration or Boolean count	Reject invalid trial/quota input rather than compute misleading compliance

Policy or trial probe	Expected interpretation
A trial fits declared resource values	Does not establish correct algorithm, truthful output or real isolation
Local permission/quota functions pass	No real path/symlink resolution, process stopping, atomic queue or escape testing performed

Guided explanation and repair

- A report omits this qualification: Resource limits constrain execution but cannot prove generated claims or algorithms correct. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise identifying the specific violated constraint

> Write operation/resource, time, output or queue rather than only safe/unsafe.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example "**Sandbox**ing and resource limits". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual correctness-limit check

- Use the worked case for **Sandbox**ing and resource limits. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured independent-limits response

- Check A against both resource limits and B against network isolation. Why must A stop after three seconds even if its **output** remains only 2 MB?

> **My explanation using a pictured step and the given facts**

Reserve the independent queued 9/proposed 2 case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete policy and quota-calculation audit

> Keep local decisions distinct from actual sandbox enforcement or measured execution.

Current public-web/outside-write investigation

A job requests a public webpage and a write outside its **output** directory. Show the intermediate steps and give the conclusion with its scope.

Complete runnable abstract-scope and quota checks

```
"""Pure sandbox-policy/quota decisions on fixtures, not an actual sandbox."""
from math import isfinite

MAIN_PAIRS=frozenset({('read','fixtures'),('write','outputs')})
def permitted(operation,resource):
    # Abstract resource handles, not filesystem paths or network addresses.
    return type(operation) is str and type(resource) is str and (operation,resource) in
MAIN_PAIRS

def admit_batch(queued,additional,maximum=10):
    if any(type(v) is not int or v<0 for v in [queued,additional,maximum]) or
queued>maximum:
        raise ValueError('invalid queue counts')
    projected=queued+additional
    return {'admit':projected<=maximum,'current':queued,'additional':additional,
            'projected':projected,'maximum':maximum,'policy':'all-or-nothing'}

def audit_picture(elapsed_s,output_bytes,network_attempted):
    if (type(elapsed_s) not in (int,float) or not isfinite(elapsed_s) or elapsed_s<0
        or type(output_bytes) is not int or output_bytes<0 or type(network_attempted) is
not bool):
        raise ValueError('invalid supplied trial measurements')
    violations=[]
    if network_attempted:violations.append('network prohibited')
    if elapsed_s>2:violations.append('time limit exceeded')
    if output_bytes>20_000_000:violations.append('output limit exceeded')
    return {'within_declared_limits':not violations,'violations':violations}

requests=[('read','fixtures'),('write','outputs'),('network','public_web'),
          ('write','outside'),('write','fixtures')]
assert [permitted(op,res) for op,res in requests]==[True,True,False,False,False]
print('Main abstract operation/resource decisions:',[(op,res,permitted(op,res)) for op,res
```

```
in requests])
assert admit_batch(8,2)['admit'] and not admit_batch(8,3)['admit']
print('Main queue admission fixtures:',admit_batch(8,2),admit_batch(8,3))
a=audit_picture(1,2_000_000,False)
b=audit_picture(1,2_000_000,True)
slow=audit_picture(3,2_000_000,False)
assert a['within_declared_limits'] and b['violations']==['network prohibited']
assert slow['violations']==['time limit exceeded']
assert audit_picture(2,20_000_000,False)['within_declared_limits']
assert audit_picture(2,20_000_001,False)['violations']==['output limit exceeded']
print('Separate picture A/B/changed A:',a,b,slow)
print('Unit convention: 1 MB = 1,000,000 bytes; 2 seconds and 20 MB are inclusive
maxima.')
print('No job is executed, queued, stopped or isolated. No file/network operation or timer
is used.')
print('These checks do not resolve paths/symlinks, atomically reserve queue space, or
prove isolation or output truth.')
```

Expected decisions, units and missing enforcement guarantees

1. Predict each policy result before running or tracing. The program uses abstract resource handles, not paths; it never opens a file, contacts a website or executes a job. It checks the main operation pairs and all-or-nothing queue fixtures, then the separate picture's supplied time/output/network records. It tests equality and one-byte-over output boundaries under the declared decimal MB convention.

   It cannot implement OS isolation, stop a process, atomically reserve queue slots, resolve symlinks or measure runtime. Treat a prohibited network proposal as a required deny decision, not proof it was actually sent. Real enforcement evidence is a separate requirement from this local policy model.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: check a new queue-admission boundary

> Apply the fixed maximum and explain what happens to the proposed batch.

Complete independent queued-task case

Queue currently has 9 tasks; job proposes 2 more under a maximum 10 policy. Decide.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Queue currently has 9 tasks; job proposes 2 more under a maximum 10 policy. Decide.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save permissions, quotas and evidence of enforcement separately

> Keep declared resource classes, admission rules, trial records and remaining implementation work.

Project task and artifact

Add a labelled **sandbox**ing and resource limits **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **sandbox**ing and resource limits.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a specific limit and a bounded policy claim

> Resource compliance cannot prove truthfulness, algorithm quality or complete isolation.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **sandbox**ing and resource limits

> **My deciding step, evidence and one remaining question**
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Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Sandbox	An enforced restriction on process capabilities
Quota	A maximum permitted resource allocation
Admission policy	Rules for accepting work under limits

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Resource limits constrain execution but cannot prove generated claims or algorithms correct. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Sandboxing and resource limits — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A job requests a public webpage and a write outside its output directory.	The fictional job may read its fixture directory, write its output directory, use no network and run for at most 2 seconds with at most 10 queued tasks. Reject excess work before starting it.	Both operations violate the sandbox contract. Deny them; public availability does not grant network or filesystem authority.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Bound a student program. A toy **sandbox** forbids network access, allows at most two seconds and limits **output** to 20 MB.

By the end, I can	How I will show it
Explain the mechanism and terminology of sandboxing and resource limits	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Bound a student program. A toy **sandbox** forbids network access, allows at most two seconds and limits **output** to 20 MB.
- Which labelled step matches this detail? Stop the prohibited network action regardless of its purpose.
- 1. Deny B network call
- 2. Apply isolation
- 3. Check A
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Deny B network call Stop the prohibited network action regardless of its purpose. **Sandbox** controls bound reach and resource use.

Passing an **output** limit cannot cancel a network or time restriction.

Changed-case question and optional hint

- Changed-case practice: A takes three seconds but still writes 2 MB. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Stop A for exceeding the time allowance.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check after the changed attempt

After your attempt, compare with the supported explanation: Stop A for exceeding the time allowance. The independent two-second limit remains binding.

Repair and retain the record

Review Untrusted **input** boundaries (ai-6/18). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract.

Does a valid z0 grant record access, and which separate gate still remains?. Return to this week and explain what you changed.

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the **mechanism** and terminology of timeouts retries and cancellation.

Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?.

Return to this week and explain what you changed.

Review State machines for bounded agents (ai-6/13). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Trace A and B through the state diagram.

Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work
</details>

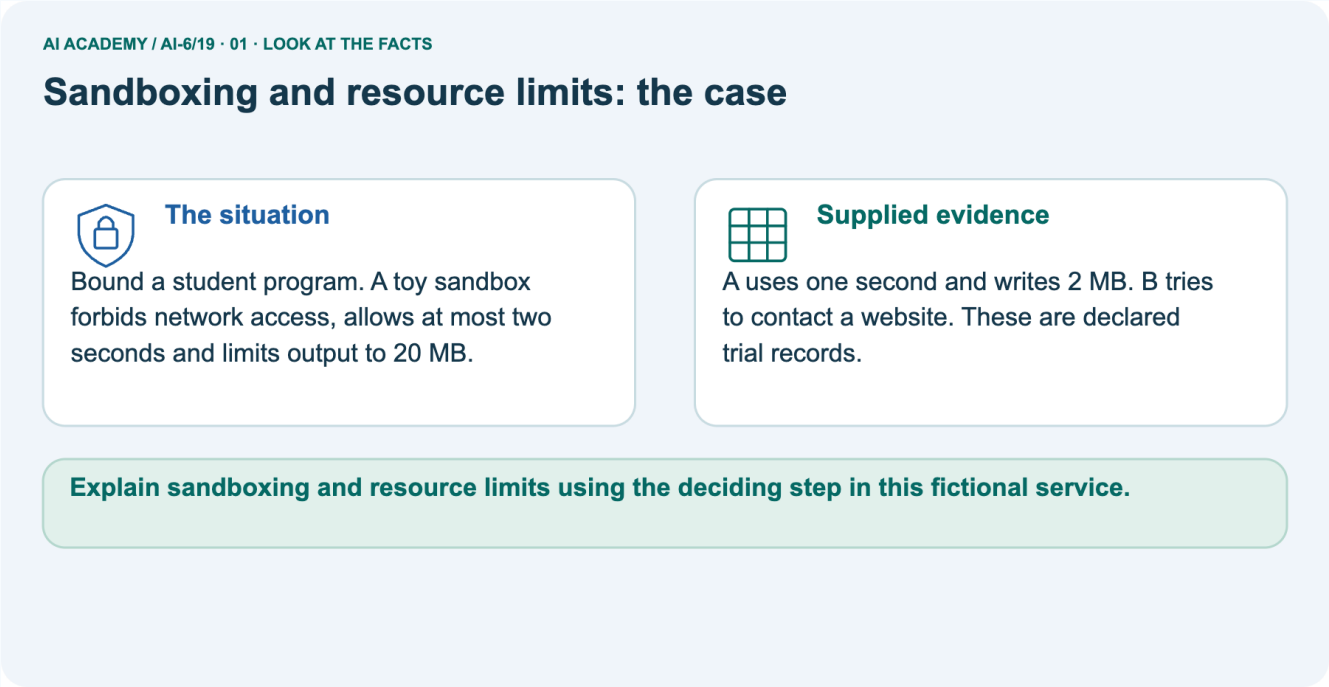
<details>
<summary>Reference · Six-picture concept journey</summary>
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REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Sandboxing and resource limits: the case



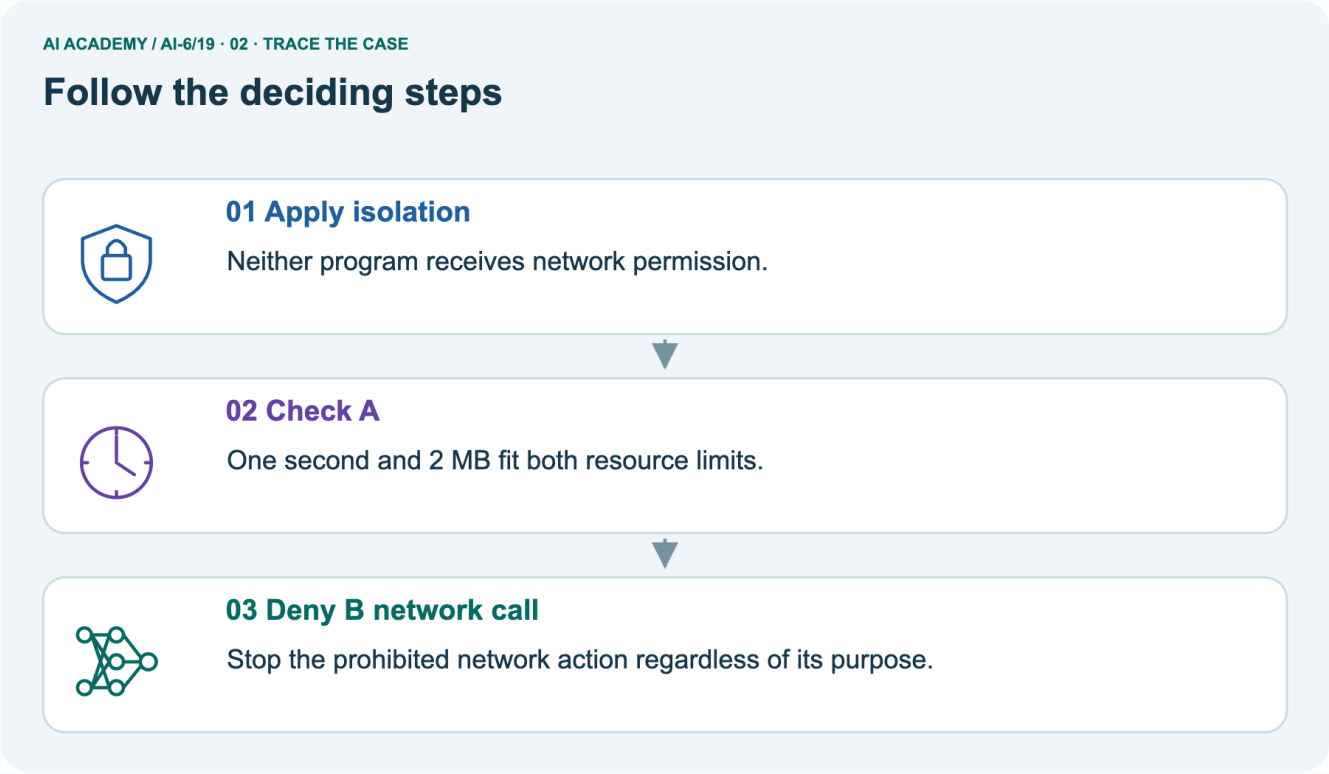
Follow the labelled picture with the matching explanation. Open the vector diagram

Bound a student program. A toy **sandbox** forbids network access, allows at most two seconds and limits **output** to 20 MB. Supplied facts: A uses one second and writes 2 MB. B tries to contact a website.

These are declared trial records.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

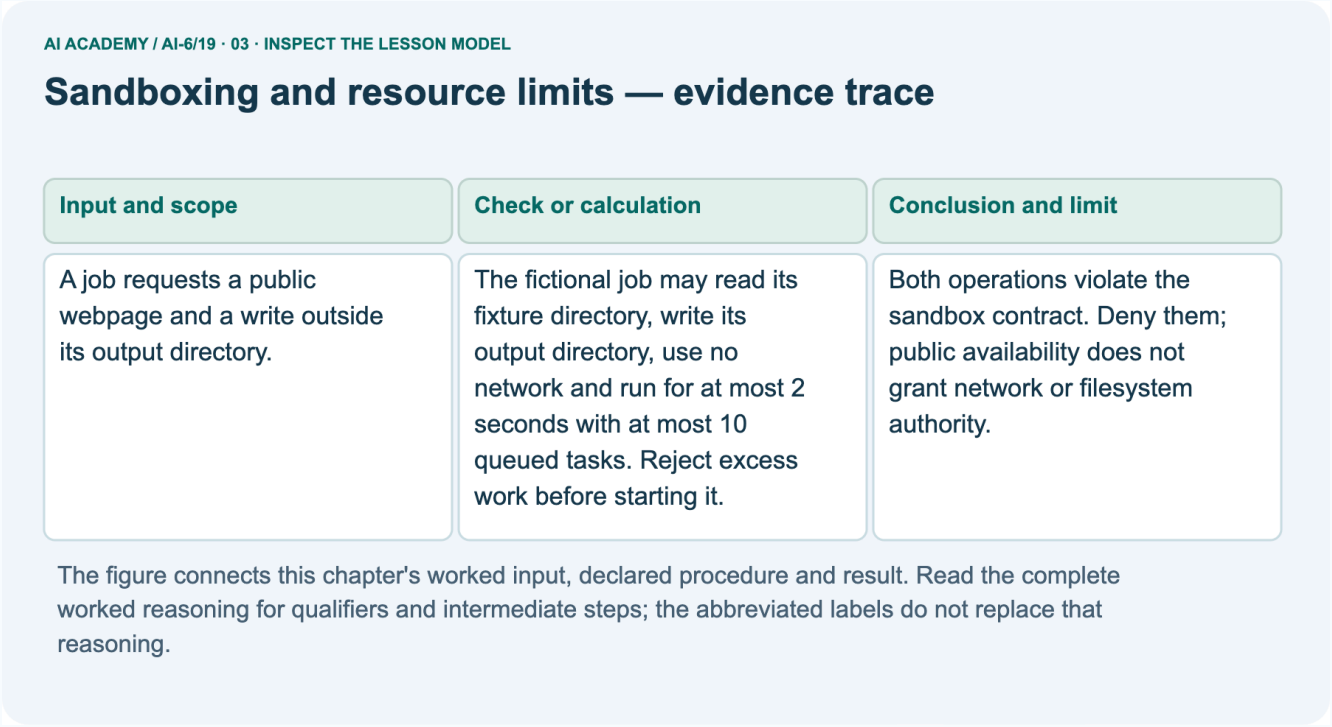


Follow the labelled picture with the matching explanation. Open the vector diagram

Apply isolation: Neither program receives network permission. → Check A: One second and 2 MB fit both resource limits. → Deny B network call: Stop the prohibited network action regardless of its purpose.

Point to each numbered stage and explain which recorded **input** it uses.

03. Sandboxing and resource limits — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

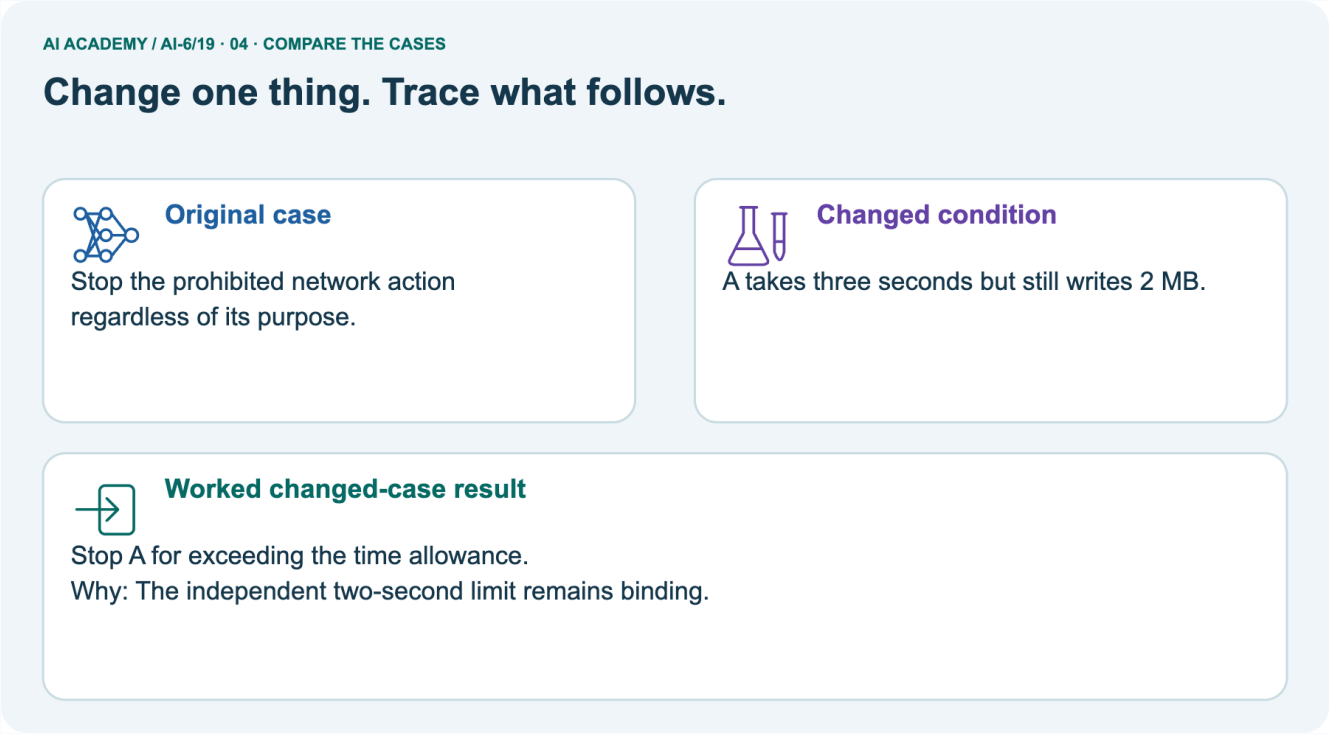
Sandboxing and resource limits — **evidence** trace.

A job requests a public webpage and a write outside its **output** directory. | The fictional job may read its fixture directory, write its **output** directory, use no network and run for at most 2 seconds with at most 10 queued tasks.

Reject excess work before starting it. | Both operations violate the **sandbox** contract. Deny them; public availability does not grant network or filesystem authority.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



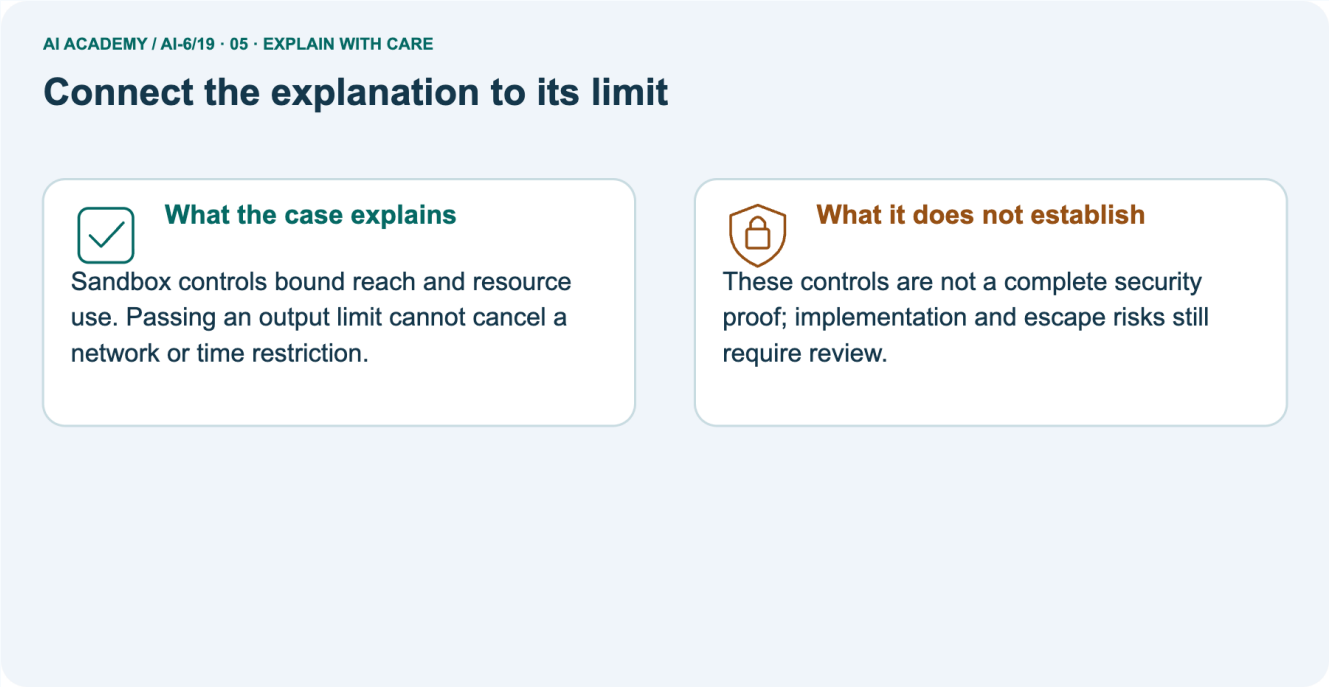
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Stop the prohibited network action regardless of its purpose. Changed condition: A takes three seconds but still writes 2 MB. Expected result: Stop A for exceeding the time allowance.

Reason: The independent two-second limit remains binding.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit

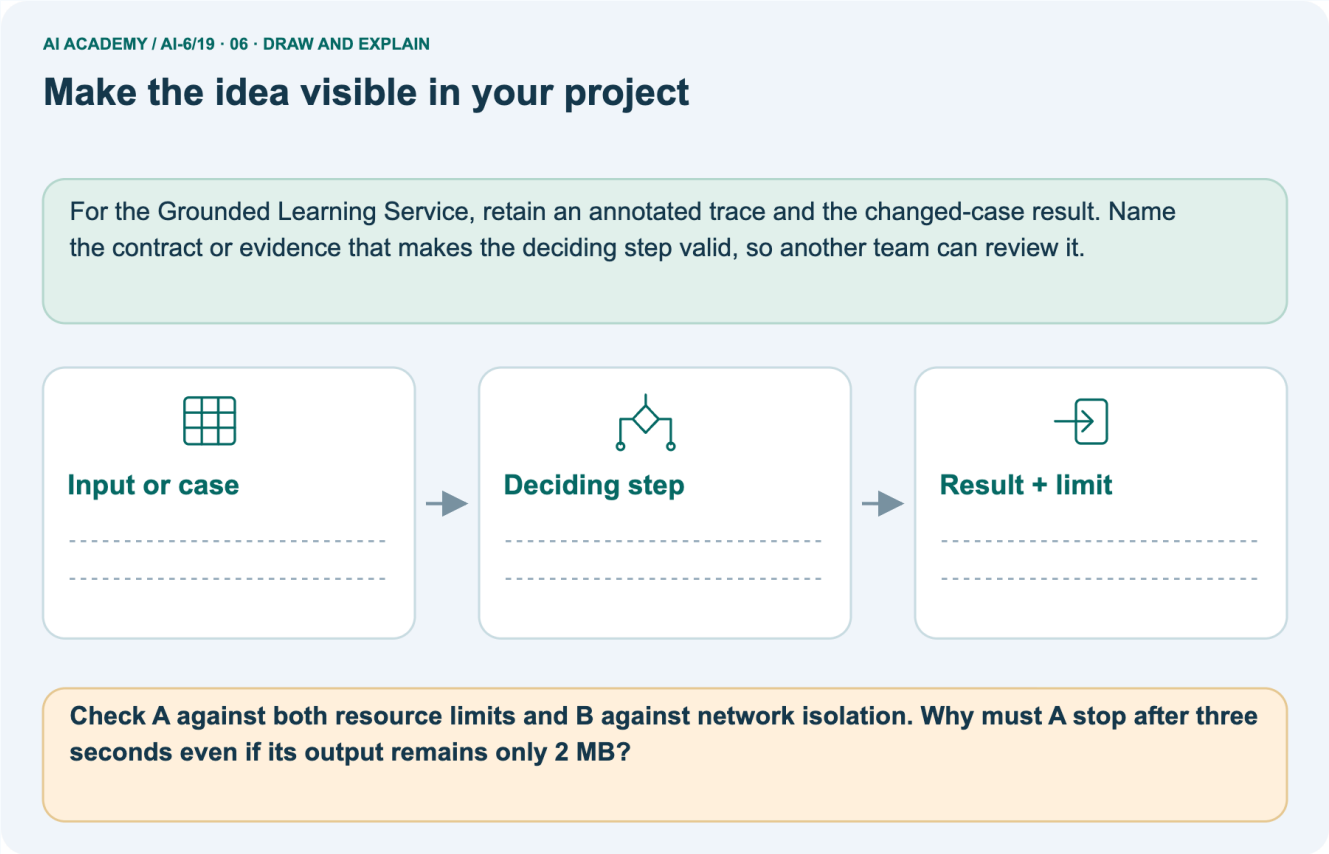


Follow the labelled picture with the matching explanation. Open the vector diagram

Sandbox controls bound reach and resource use. Passing an **output** limit cannot cancel a network or time restriction. Limit: These controls are not a complete security proof; implementation and escape risks still require review.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Check A against both resource limits and B against network isolation. Why must A stop after three seconds even if its **output** remains only 2 MB?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Sandboxing and resource limits: the case

02. Follow the deciding steps

03. Sandboxing and resource limits — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.